EMERGENCY FLIGHT REBOOKING SYSTEM

A MINI PROJECT REPORT

IT2304 – FULL STACK WEB DEVELOPMENT

Submitted by

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DEPARTMENT OF INFORMATION TECHNOLOGY

KAMARAJ COLLEGE OF ENGINEERING AND TECHNOLOGY

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BONAFIDE CERTIFICATE

Certified that the project report "**EMERGENCY FLIGHT REBOOKING SYSTEM**" is the bonafide work of " **N. GOKULAPRIYA (920424205027) & K. RAJASHREE (920424205078) & R. SAKTHI SHREE (920424205086)"** who carried out the mini project work as a part of **IT2304 – Full Stack Web Development** under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

CHAPTER 1

INTRODUCTION

1.1 MERN Stack

MERN Stack is a full-stack web development technology that uses JavaScript to build modern applications. MERN represents **MongoDB, Express.js, React, and Node.js**. These technologies work together to develop the user interface, handle server-side operations, and store application data. In the Emergency Flight Rebooking System, MERN Stack is used to manage flight information and passenger rebooking requests.

1.2 Importance of MERN Stack

MERN Stack makes web application development easier by using JavaScript throughout the application. It supports efficient data processing, interactive user interfaces, and smooth communication between frontend and backend components. It also helps developers maintain application code and extend features when required. These benefits make MERN Stack suitable for developing the Emergency Flight Rebooking System.

1.3 Basic Terminologies

• **MongoDB:** A NoSQL database used to store passenger details, flight information, booking records, and rebooking requests.

• **Express.js:** A backend framework for Node.js used to handle application requests and create REST APIs.

• **React:** A JavaScript library used to develop interactive pages and dashboards for passengers and administrators.

• **Node.js:** A JavaScript runtime environment used to execute server-side code and process application requests.

1.4 Components of MERN Stack

1.4.1 MongoDB

MongoDB is a document-oriented NoSQL database that stores information in flexible formats. In this project, it manages passenger accounts, flight schedules, booking details, cancellation records, and rebooking requests.

1.4.2 Express.js

Express.js is a web framework that simplifies backend development using Node.js. It handles HTTP requests and provides REST APIs for communication between the frontend and database. In this project, it processes flight-related operations and passenger rebooking requests.

1.4.3 React

React is a JavaScript library used to build interactive user interfaces with reusable components. In this project, it provides pages for searching flights, submitting rebooking requests, checking request status, and managing flight information through the admin dashboard.

1.4.4 Node.js

Node.js is a JavaScript runtime that executes code on the server side. It works with Express.js to process requests, manage application logic, and communicate with MongoDB. In this project, it supports flight management and rebooking operations.

1.5 Operations in MERN Stack

Common operations include:

• **Create:** Adding flight details and submitting new rebooking requests.

• **Read:** Retrieving flight schedules, booking information, and request statuses.

• **Update:** Modifying flight details and changing rebooking request statuses.

• **Delete:** Removing eligible flight records or outdated information when authorized.

• **API Communication:** Exchanging data between the React frontend and Express.js backend through REST APIs.

1.6 Need for MERN Stack

MERN Stack provides:

- Full-stack development using JavaScript.
- Simplified application development and maintenance.
- Efficient storage and retrieval of flight records.
- Interactive interfaces for passengers and administrators.
- Reusable React components.
- Flexible backend development.
- Smooth communication between frontend and backend.

1.7 Applications of MERN Stack

MERN Stack is used in:

- Airline booking and flight management systems.
- Online travel reservation applications.
- E-commerce platforms.
- Online food ordering systems.
- Social networking applications.
- Content management systems.
- Business management applications.
- Real-time web applications.

CHAPTER 2

METHODOLOGY

2.1 Objective

The main objective of the Emergency Flight Rebooking System is to provide an efficient and user-friendly platform for managing unexpected flight cancellations and rebooking activities. The system aims to reduce passenger inconvenience by simplifying the process of finding alternative flights, submitting rebooking requests, and tracking booking and refund information. It also helps administrators manage flight details and passenger requests effectively. The specific objectives of this project are:

- To provide instant alerts and updates about unexpected flight cancellations.
- To help passengers search for suitable alternative flights based on their travel requirements.
- To simplify the flight rebooking process through an online request system.
- To allow passengers to track their rebooking requests and refund status.
- To provide an admin dashboard for managing flights, cancellations, and passenger requests..

2.2 Problem Statement

Flight cancellations and unexpected schedule changes due to bad weather, technical issues, operational problems, and other emergencies create difficulties for passengers. Finding alternative flights, submitting rebooking requests, and tracking booking or refund status can be a time-consuming and stressful process. Traditional rebooking methods often involve manual communication, long waiting times, and a lack of proper updates. Therefore, there is a need for a centralized and user-friendly system that simplifies emergency flight rebooking. The proposed Emergency Flight Rebooking System aims to overcome these challenges by providing cancellation alerts, alternative flight searches, online rebooking requests, and refund tracking facilities. It also enables administrators to manage flight information and passenger requests efficiently, ensuring a faster, more organized, and convenient rebooking process.

2.3 Project Description

In the current air travel system, Passengers may face difficulties when flights are cancelled or rescheduled due to bad weather, technical failures, operational issues, or other emergencies. Finding alternative flights, communicating with airline authorities, and completing the rebooking process can be time-consuming and confusing. The Emergency Flight Rebooking System provides a centralized online platform where passengers can receive cancellation alerts, search for alternative flights, submit rebooking requests, and track booking and refund status. The admin dashboard allows administrators to manage flights, cancellations, passenger requests, and rebooking approvals. By automating these activities, the system reduces manual work, saves time, improves communication, and provides a convenient rebooking experience.

Modules

Passenger Functionalities

- **User Registration and Login**: Passengers can create an account and log in securely to access the rebooking system.
- **Flight Information**: Passengers can view flight details, schedules, and current flight status.
- **Flight Cancellation Alerts**: Passengers can view cancellation information and details about affected flights.
- **Alternative Flight Search**: Passengers can search for alternative flights based on departure location, destination, and travel date.
- **Flight Details**: Passengers can view alternative flight information, including flight number, schedule, and availability.
- **Rebooking Request**: Passengers can select an alternative flight and submit a rebooking request.
- **Request Tracking**: Passengers can track request statuses such as Pending, Approved, or Rejected.
- **Booking Details**: Passengers can view their original flight details and rebooking information.
- **Refund Status**: Passengers can check the status of refund requests for cancelled flights, where applicable.

- **Rebooking History**: Passengers can view their previous rebooking requests and their outcomes.

Admin Functionalities

- **Admin Dashboard**: Administrators can view flight details, cancellation records, passenger requests, and rebooking statistics.
- **Flight Management**: Administrators can add, edit, and manage flight information.
- **Cancellation Management**: Administrators can update flight statuses and manage cancellation details.
- **Alternative Flight Management**: Administrators can manage available alternative flights for affected passengers.
- **Rebooking Request Management**: Administrators can view and review passenger rebooking requests.
- **Request Approval and Rejection**: Administrators can approve or reject requests based on alternative flight availability.
- **Passenger Management**: Administrators can view and manage passenger records.
- **Refund Management**: Administrators can update and monitor refund records, where applicable.
- **Booking Management**: Administrators can manage original booking details and rebooking records.

CHAPTER 3

RESULTS AND DISCUSSION

Github Link : https://github.com/gokula628/MERNSTACK_SkyRebook.git

Screenshots

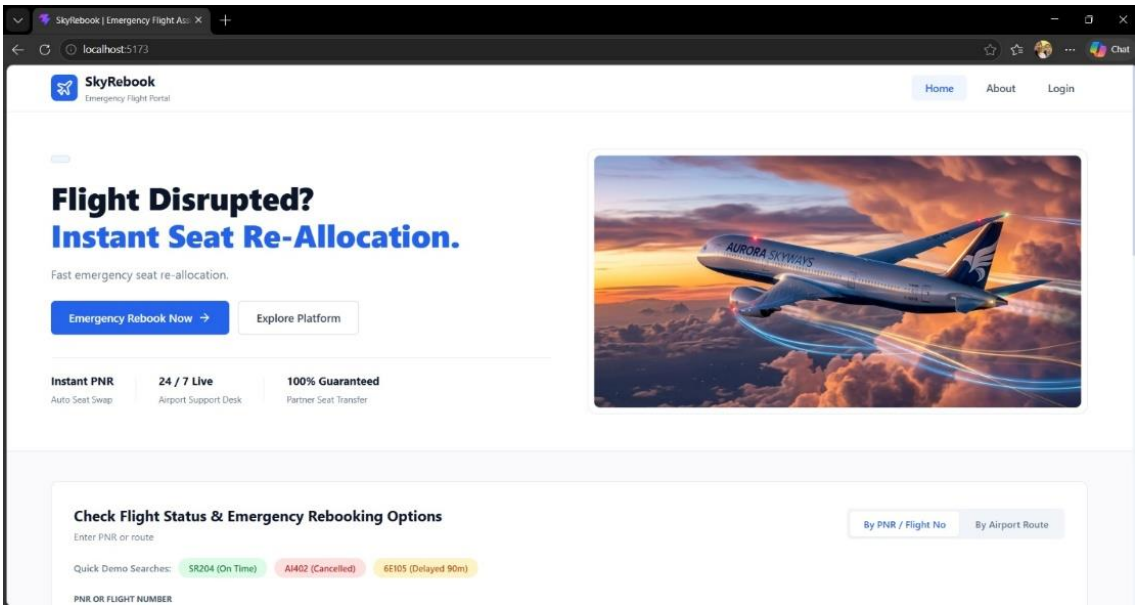


Figure 1: Emergency Flight Rebooking System – Home Page and Flight Search Interface

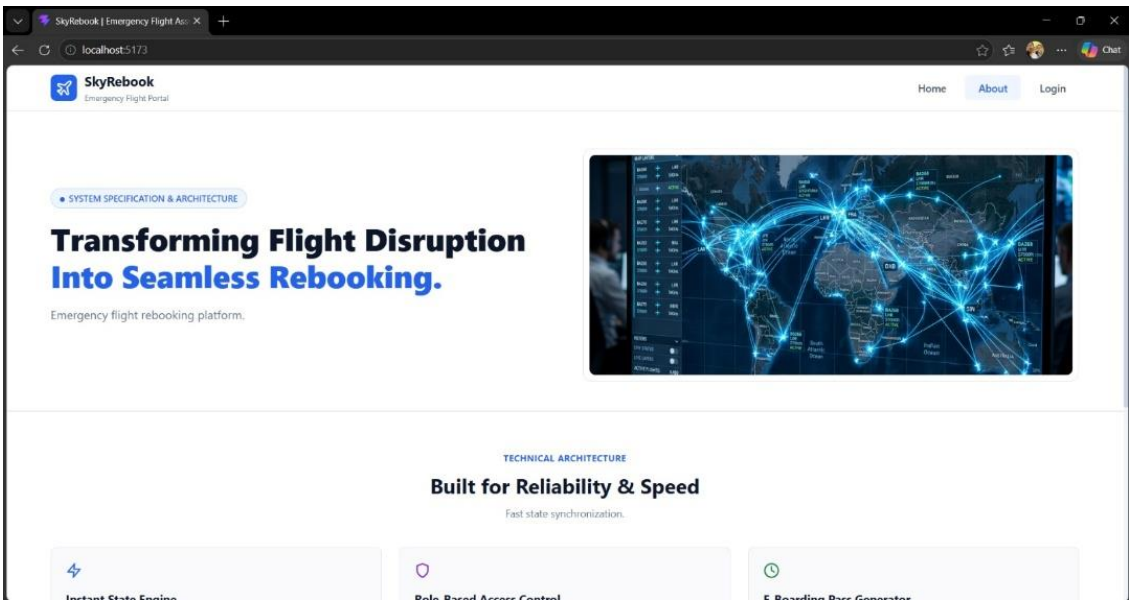


Figure 2: System Specification and Technical Architecture of the Emergency Flight Rebooking System

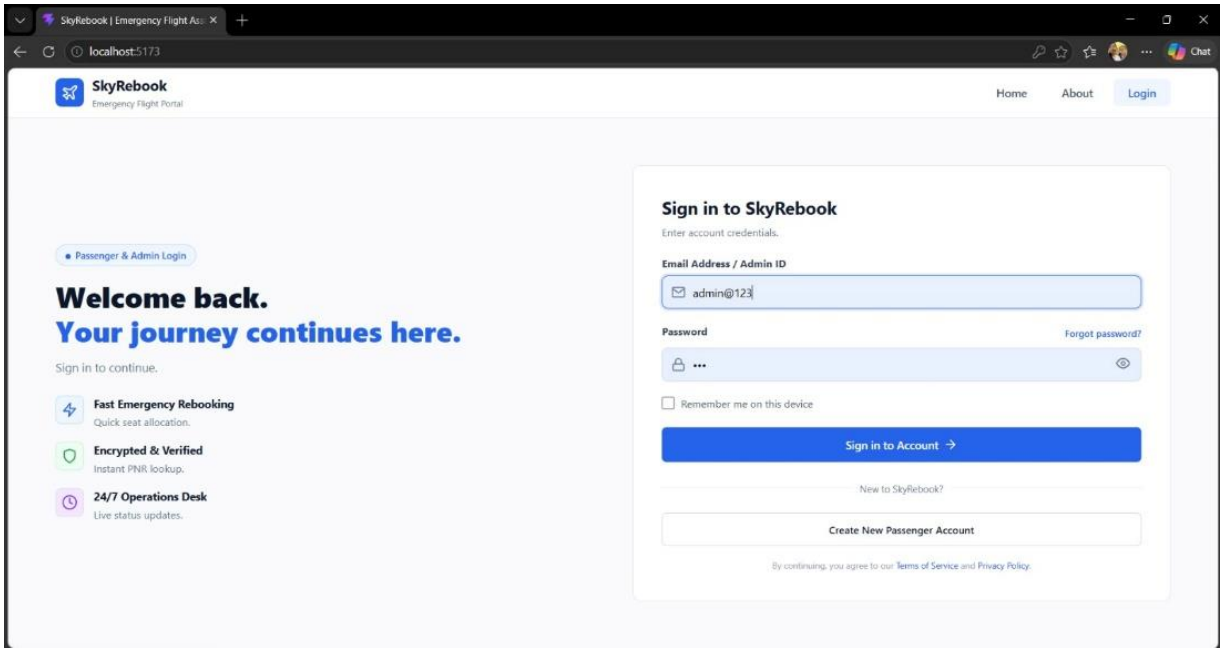


Figure 3: User and Administrator Login Interface of the Emergency Flight Rebooking System

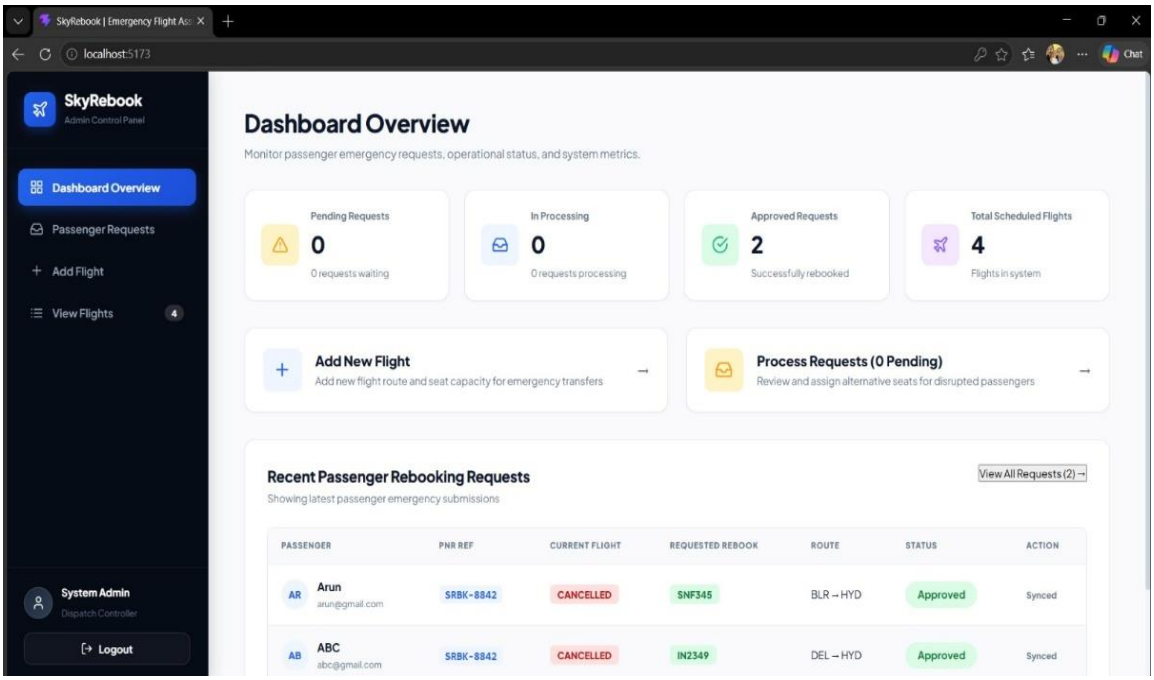


Figure 4: SkyRebook Admin Dashboard – Overview of Emergency Rebooking Requests and Flight Operations

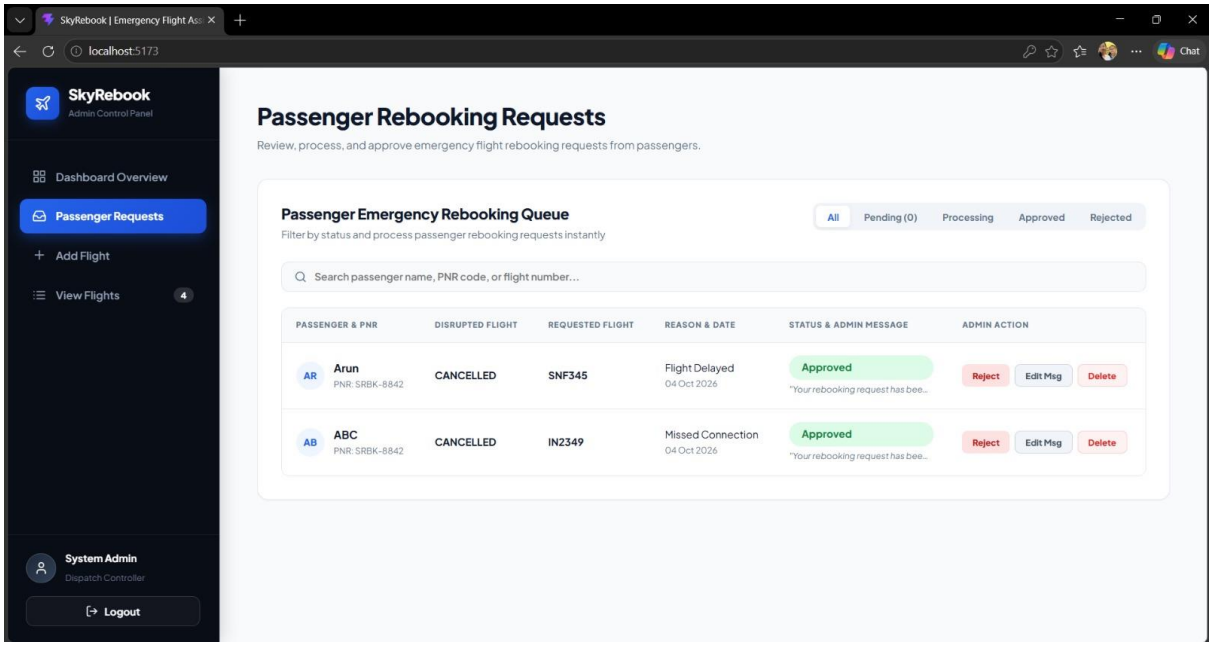


Figure 5: SkyRebook Passenger Rebooking Queue – Management of Emergency Flight Rebooking Requests

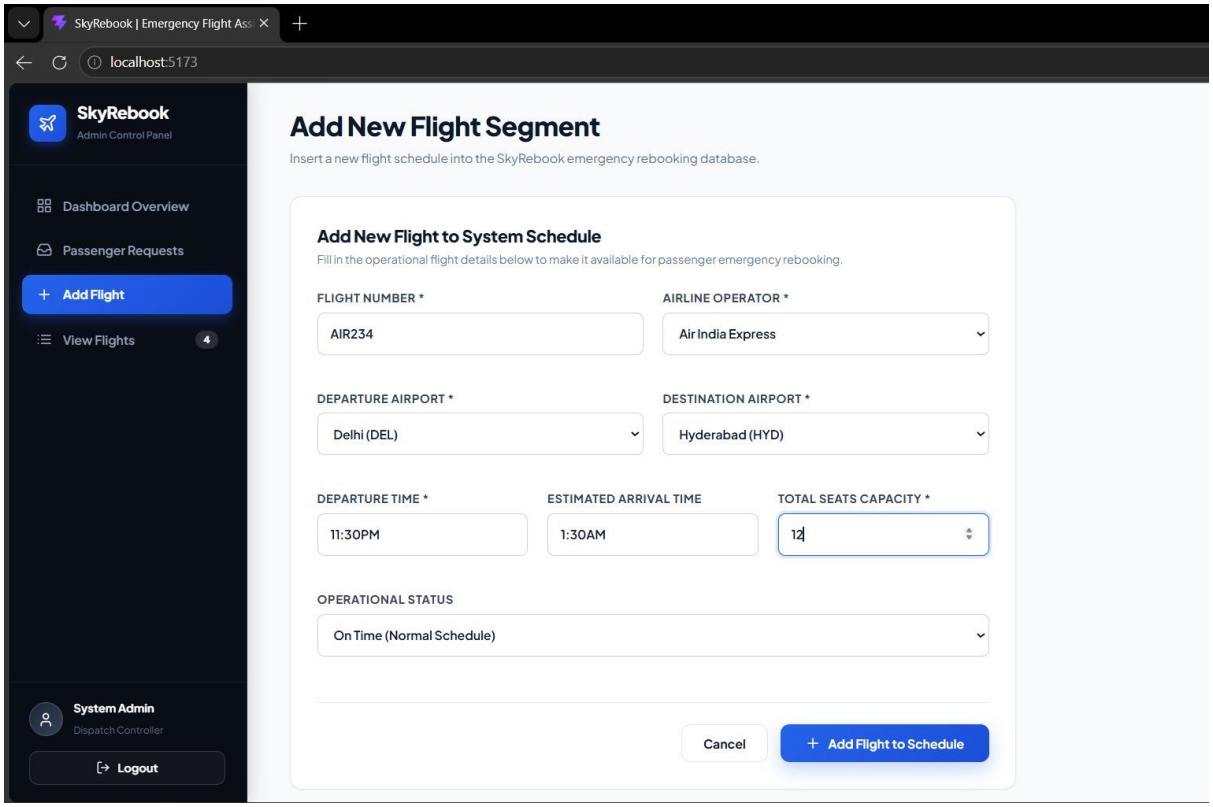


Figure 6 : SkyRebook Add Flight Segment – Entry of New Flight Schedule and Operational Details

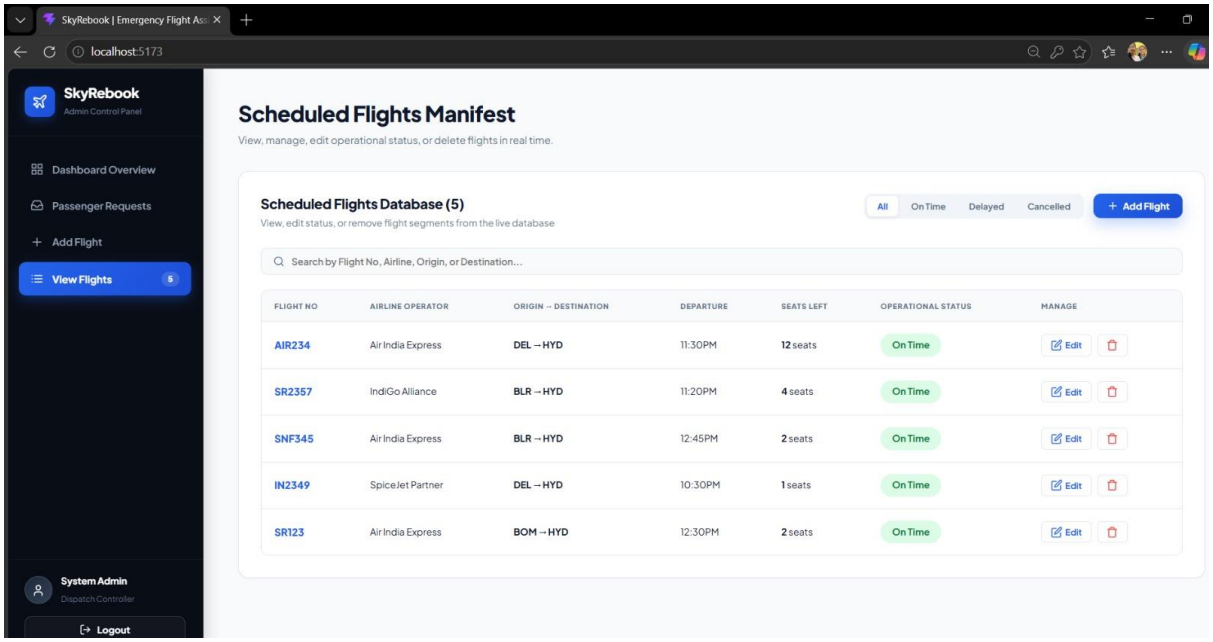


Figure 7: SkyRebook Scheduled Flights Manifest – Overview of Available Flight Schedules and Operational Status

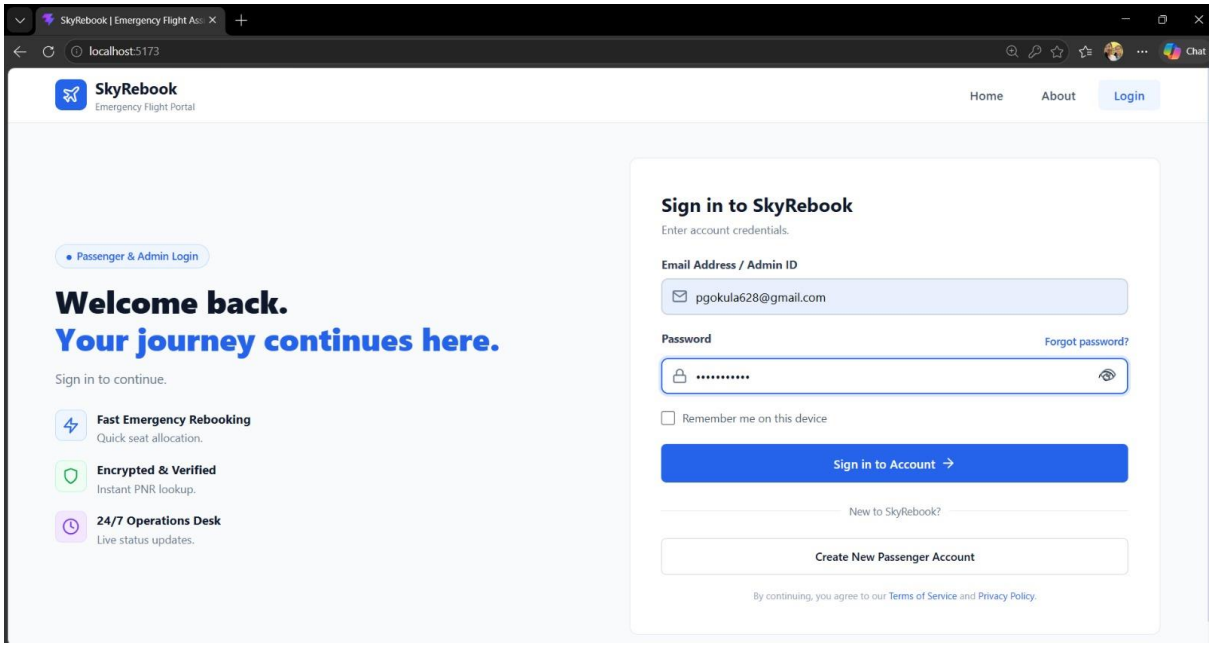


Figure 8: SkyRebook Login Page – Passenger and Administrator Account Authentication

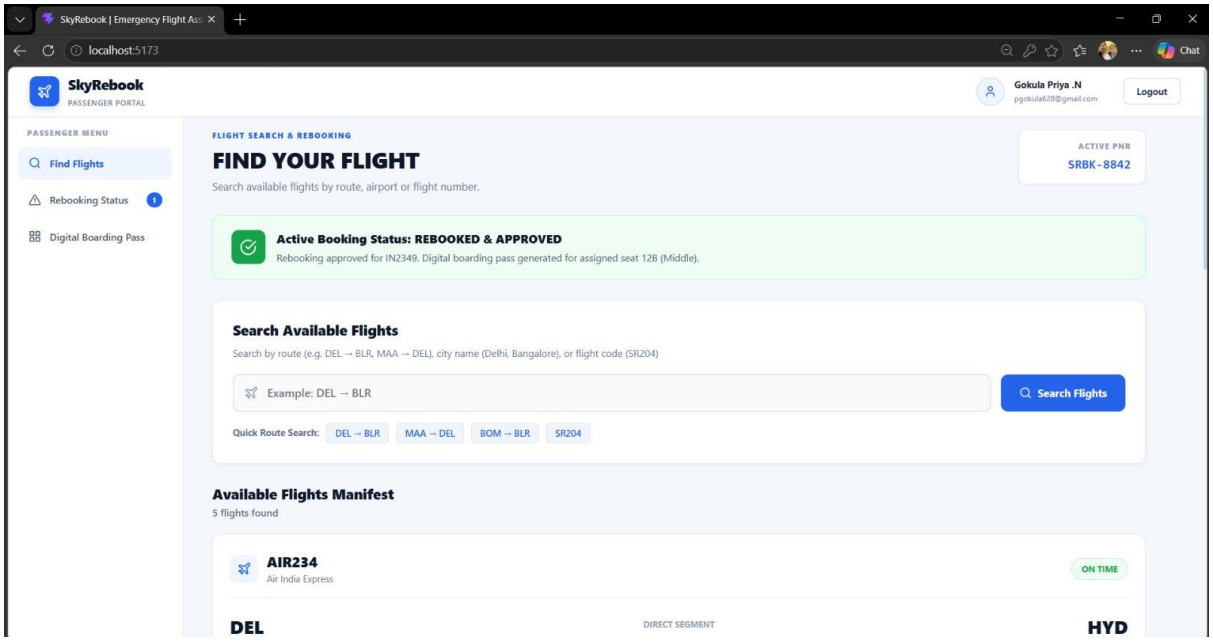


Figure 9: SkyRebook Passenger Portal – Flight Search and Emergency Rebooking Status

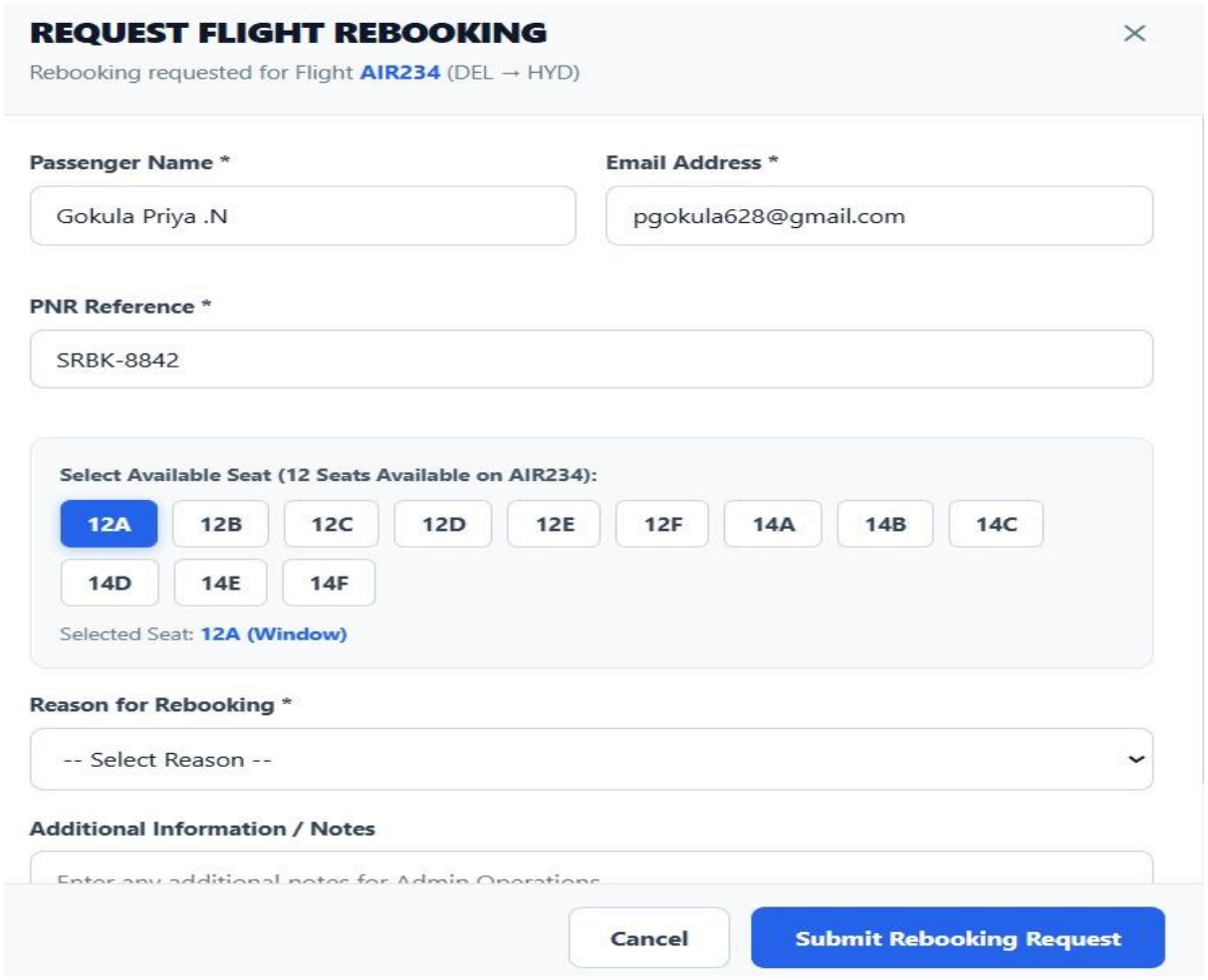


Figure 10: SkyRebook Flight Rebooking Request – Passenger Seat Selection and Rebooking Details

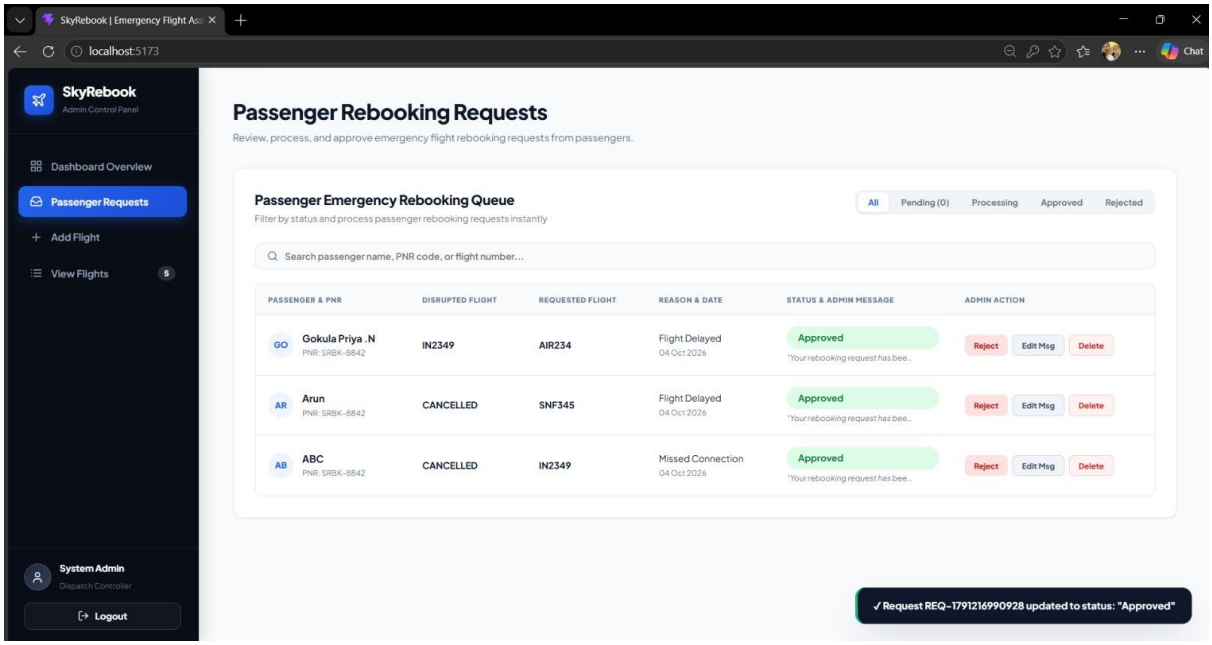


Figure 11: SkyRebook Admin Rebooking Queue – Approval and Management of Passenger Requests

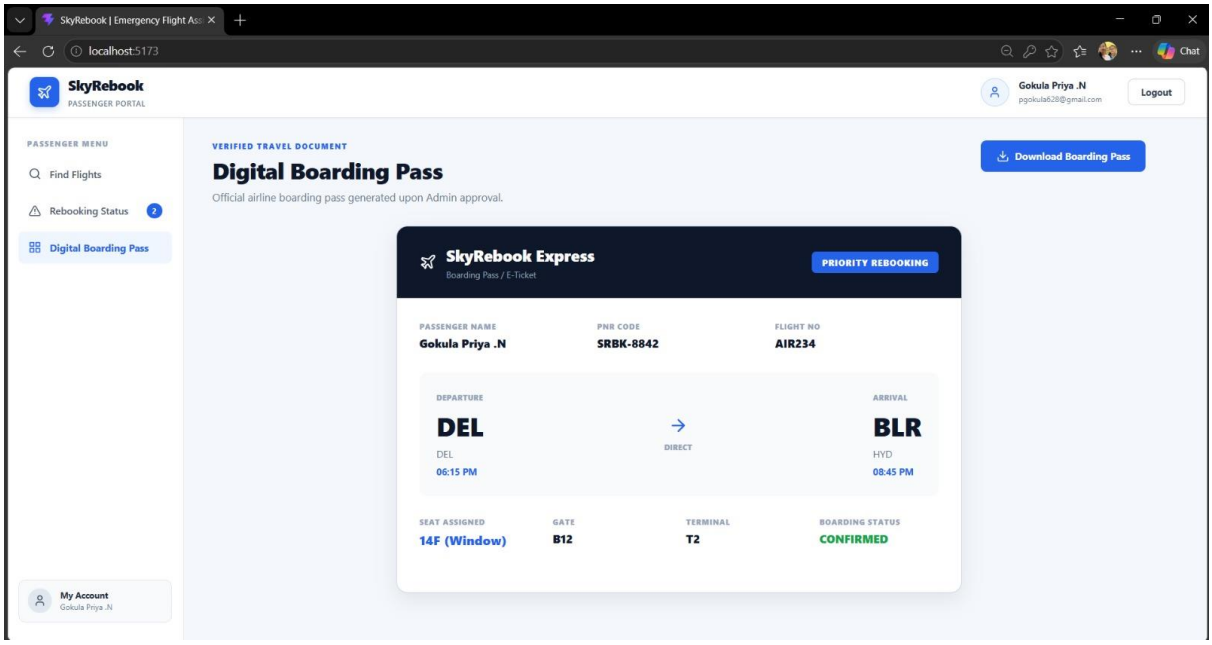


Figure 12: SkyRebook Digital Boarding Pass – Confirmed Flight and Passenger Travel Detail

```
_id: ObjectId('6ac22551276ffdc701e8e3e5')
requestId: "REQ-1791108433186"
pnr: "SRBK-8842"
passengerName: "Gokula Priya .N"
email: "pgokula628@gmail.com"
currentFlight: "CANCELLED"
requestedFlight: "IN2349"
requestedSeat: "12B (Middle)"
route: "DEL → HYD"
reason: "Flight Delayed"
message: "Passenger requested alternative flight allocation."
createdAt: ISODate('2026-10-04T10:07:13.209+00:00') October 4, 2026 at 10:07:13 AM UTC
updatedAt: ISODate('2026-10-04T10:07:13.209+00:00') October 4, 2026 at 10:07:13 AM UTC
__v: 0
```

```
_id: ObjectId('6ac226ff4fe73f3e8f576b6e')
requestId: "REQ-1791108863281"
pnr: "SRBK-8842"
passengerName: "ABC"
email: "abc@gmail.com"
currentFlight: "CANCELLED"
requestedFlight: "IN2349"
requestedSeat: "12A (Window)"
route: "DEL → HYD"
reason: "Missed Connection"
message: "Passenger requested alternative flight allocation."
createdAt: ISODate('2026-10-04T10:14:23.282+00:00') October 4, 2026 at 10:14:23 AM UTC
updatedAt: ISODate('2026-10-04T10:14:23.282+00:00') October 4, 2026 at 10:14:23 AM UTC
__v: 0
```


Figure 13: Rebooking Requests Submitted by Users.


```
_id: ObjectId('6ac226ff4fe73f3e8f576b6f')
requestId: "REQ-1791108863281"
pnr: "SRBK-8842"
passengerName: "ABC"
email: "abc@gmail.com"
currentFlight: "CANCELLED"
requestedFlight: "IN2349"
requestedSeat: "12A (Window)"
route: "DEL → HYD"
reason: "Missed Connection"
message: "Passenger requested alternative flight allocation."
createdAt: ISODate('2026-10-04T10:14:23.284+00:00') October 4, 2026 at 10:14:23 AM UTC
updatedAt: ISODate('2026-10-04T10:14:23.284+00:00') October 4, 2026 at 10:14:23 AM UTC
__v: 0
```

```
_id: ObjectId('6ac3b36cb5d1b53818517341')
requestId: "REQ-1791210348071"
pnr: "SRBK-8842"
passengerName: "Arun"
email: "arun@gmail.com"
currentFlight: "CANCELLED"
requestedFlight: "SNF345"
requestedSeat: "12C (Aisle)"
route: "BLR → HYD"
reason: "Flight Delayed"
message: "afdgryry"
createdAt: ISODate('2026-10-05T14:25:48.101+00:00') October 5, 2026 at 2:25:48 PM UTC
updatedAt: ISODate('2026-10-05T14:25:48.101+00:00') October 5, 2026 at 2:25:48 PM UTC
__v: 0
```


Figure 14:MongoDB- User Rebooking Requests Received by Admin.

CHAPTER 4

CONCLUSION

The Emergency Flight Rebooking System provides an efficient, reliable, and user-friendly solution for managing unexpected flight cancellations and schedule changes. The system is developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js), which helps create a responsive interface, handle server-side operations, and manage passenger and flight information effectively. It simplifies the rebooking process by allowing passengers to receive cancellation alerts, search for suitable alternative flights, submit rebooking requests, and track their booking and refund status without facing unnecessary difficulties. It also provides an admin dashboard through which administrators can manage flight details, cancellation information, passenger records, and rebooking approvals in an organized manner. This application reduces manual work, saves time, minimizes communication gaps, and helps passengers make quick decisions during unexpected travel disruptions. It also improves coordination between passengers and administrators by providing a centralized platform for managing rebooking activities. Overall, the proposed system aims to make emergency flight management faster, easier, and more convenient while improving the passenger travel experience. In the future, the system can be enhanced by integrating intelligent flight recommendations, online payment facilities, real-time notifications, and mobile application support to provide more advanced services and improve the overall efficiency of the application.

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